

Outline the clinical criteria for diagnosis of brainstem death. Outline the physiological implications of brain stem death following SAH in a patient listed for organ donation.

(Modified from HKCA 2012, FANZCA 2016A)

Examiner Report

Outline the clinical criteria for diagnosis of brain-stem death.

(11/20 passed)

This question asked for clinical criteria for diagnosis of brain-stem death. Good candidates were able to stratify the criteria as preconditions, exclusions and clinical tests with brief description. In general, apnoea tests were not well answered. Candidates who failed did not understand the principles of the tests e.g. to avoid damage to the brain during apnoea test, it should not be performed until cranial nerves examination had demonstrated brainstem areflexia

Outline the physiological implications of brainstem death following subarachnoid haemorrhage in a patient listed for organ donation.

Borderline standard: Recognises key potential issues that are likely to be needed to be managed if patient is going to provide suitable organs for donation:

1. inadequate perfusion of vital organs
2. some aspect of hypothalamic dysfunction
3. the physiological effect of the increase in intra-cranial pressure

Comment: The standard of performance for this question was extremely poor with most candidates not answering the question correctly. The question needed candidates to understand the physiological changes that occur with brain stem death and then to relate this to maintaining organ survival in order to undergo donation. Many candidates provided information on brain stem death assessment, liaising with relatives, consent, etc. with resultant low scores. It is important to read the question very carefully and understand what is being asked.

Sample Answer

There are several pre-conditions required before brainstem death certification.

1. Presence of objective proof (CT or MRI) of sufficient irreversible structural brain damage, leading to total and irreversible cessation of brainstem functions.
2. Neurological depression or coma secondary to CNS depressant drugs, sedative drugs, hypothermia, metabolic and endocrine causes must be excluded
 - Na 125-155, K 3-5.5, glucose 4-15, SBP>100, SpO₂ >95%, T >35C, no clinical evidence of renal or hepatic failure
3. Presence of intact neuromuscular function

Examination should be done by 2 independent consultants not involved in direct patient care:

Brainstem Test	Description of test	Expected response if brain dead
Pupillary Light Reflex	Shine strong light into each eye	Both pupils fixed and non-reactive to strong light
Corneal Reflex	Touch cornea using a wisp of cotton wool or moistened cotton bud	No muscle contractions noted around the eyes
Oculocephalic (Doll's eye) Reflex	Turn the head quickly 90 degrees to the left and then to the right	Eyes should remain fixed to the midline and move with the head
Response to Pain Stimulation	Apply pain stimuli centrally e.g. at the supra-orbital ridge and peripherally to the nail beds	No motor response seen anywhere to a centrally applied pain stimulus AND no facial response seen to a peripherally applied pain stimulus
Gag and Cough Reflex	Pass suction catheter down endotracheal tube to level of carina and apply firm pressure to posterior pharynx with wooden spatula	No gag or cough response elicited
Cold Caloric Test	Inject 50-100 ml of cold water into each ear slowly over 1 min with head of bed elevated to 30 degrees (wait 5 min before testing other side)	No tonic deviation of pupil towards the side being syringed should be seen
Apnoea Test	Pre-oxygenate for 10 min followed by insufflation of 6 L/min of O ₂ via suction catheter into endotracheal tube while disconnected from ventilator for 6-8 min	No spontaneous breathing efforts seen at PaCO ₂ >50 mmHg (in patients with chronically elevated PaCO ₂ levels, use an elevation of >20 mmHg above baseline as the threshold)

Apnoea test should be done only after cranial nerve examination demonstrated brainstem areflexia to avoid damage to an intact, functioning brain.

If any of the abovementioned tests cannot be performed or uncertainty over significance of these responses, then supplementary tests may be performed. Cerebral angiography to confirm that absent intracranial blood flow and radionuclide scan to confirm absent intracranial perfusion.

Physiological issues which may affect viability of organs for donation in a patient with brain stem death following SAH include:

1. Impaired perfusion of vital organs
2. Hypothalamic dysfunction
3. Effects from raised intracranial pressure (from SAH)

Cardiovascular

- Compensatory hypertension may be seen initially secondary to Cushing's Reflex to maintain cerebral perfusion pressure (CPP). This is usually temporary.
- Hypotension *should be corrected promptly to maintain perfusion pressures to heart, liver and kidneys.*
 - May be secondary to reduced cardiac contractility, peripheral vasodilatation (loss of function of vasomotor centre, DVC), and fluid losses (diabetes insipidus).
 - *Fluids and vasopressors should be started.*
 - *Anti-hypertensives are typically stopped unless persistent MAP > 95 mmHg.*
 - *Combined hormone replacement (thyroid hormones, glucocorticosteroids, and desmopressin) may be needed if there is persistent haemodynamic instability.*
 - *Invasive BP monitoring with SBP >100mmHg, MAP >70 mmHg*
 - *HR 60-120 BPM*
- Cardiac arrhythmias *should be treated aggressively*
 - Tachy-arrhythmias *may be treated with amiodarone and beta blockers.*
 - Brady-arrhythmias *(more likely than tachyarrhythmia) due to loss of vagal tone can be resistant to anticholinergic drugs, and may require treatment with adrenaline.*
 - *Electrolytes such as K⁺ and Mg²⁺ should be replaced if deficient.*

Respiratory

- Neurogenic pulmonary oedema
 - Secondary to 'catecholamine storm' in response to raised ICP, and worsened by systemic inflammation following brain death.
 - *Lung protective ventilation strategies with PEEP use to improve oxygenation while minimising barotrauma risk.*
 - *Avoid excessive fluid administration.*
 - *Use of IV Methylprednisolone (15mg/kg) to attenuate pulmonary oedema is controversial.*
 - Pulmonary hypertension can occur following neurogenic pulmonary oedema. *Maintain adequate oxygenation, normocapnia, normothermia.*
- Higher risks of respiratory infections. *30deg head tilt to minimise aspiration risk, with continuation of PPI. Diligent aseptic tracheal suctioning should be done. Use of PEEP to reduce atelectasis.*
 - *Lowest FiO₂ to keep SpO₂ >95%, PaO₂ >80 mmHg*
 - *PaCO₂ 35-50 mmHg*
 - *Ph 7.35 -7.45*
 - *Plateau pressure < 30 mmHg*

Endocrine and Metabolic

- Hypothalamic dysfunction (and pituitary ischemia) are common following brain death, and typically leads to loss of posterior pituitary function (ADH production) and anterior pituitary function (TSH, ACTH production)
 - Neurogenic Diabetes Insipidus (DI)/ cerebral salt wasting *should be treated by replacing fluid and electrolyte losses. DDVAP may be used.*
 - Hypothyroidism *may benefit from T3/T4 replacement, which may improve cardiac function.*
 - Hypocortisolism *may benefit from hydrocortisone replacement.*

- Hyperglycemia from insulin resistance and deficiency *should be corrected to keep patient normoglycemic.*
 - Serum glucose 4-10 mmol/L
- Maintenance of normothermia with active warming devices may be required due to impaired thermoregulatory function of the hypothalamus (with hypothermia worsened by decreased metabolism, hypothyroidism, and vasodilation)
 - Core temperature 35-38C

Renal and Electrolytes

- Renal perfusion may be impaired, *and should be maintained with careful assessment of fluid status, and monitoring of urine output to avoid over-/under-hydration.*
 - Urine output 1-2 ml/kg/h
 - Serum Na^+ 130-150 mmol/L
 - Normal K^+ , Ca^{2+} , Mg^{2+} , PO_4^-
- Electrolyte abnormalities (secondary to DI/use of hypertonic saline for treatment of raised ICP) are common, and should be corrected
 - Avoid hypernatremia ($\text{Na} > 155$), which can lead to impaired graft function.
- Nephrotoxic drugs should be stopped.

Liver

- Liver glycogen depletion *may be minimised by maintaining adequate nutrition.*
- Liver congestion *may be minimised by avoiding excessive fluid administration, and maintaining normal CVP < 10 mmHg.*
- Hepatotoxic drugs should be stopped.
- Similar to renal management, avoiding hypernatremia ($\text{Na} > 155$) will help to reduce risk of impaired graft function.

Haematological

- Coagulopathy from DVI/thromboplastin release *should be corrected*
 - Target INR < 1.5, platelet count > 50 000 /mm³
 - Aim Hb > 7 g/dL
- Thromboembolism risk is increased due to prolonged immobility. *Use of compression stockings and intermittent calf compressive devices may be used for prevention.*

Others

- Non-essential medications should be stopped.
- Preventive measures, early detection and prompt treatment of infections, including antibiotics administration, should be continued.
- Eye protection (eye tapes, eye drops) especially if cornea transplant is considered.